

Why fresh DE is more trustworthy than the warm_start results

| Section for Stage 1 closure / Stage 2 motivation in the progress report.

TL;DR

The warm_start data underlying the original Pareto analysis was generated by a fitness pipeline with two confirmed defects: an incomplete random-seed injection in `s4_personalize_fig7_v7.py`, and a methodologically inappropriate PAC pipeline (Hilbert-based phase + cross-correlation lag) for the model's pulse-type firing-rate signals. After fixing both — verified by bit-identical reproducibility on the 5 Pareto seeds — we re-ran a fresh, deterministic Sobol-initialised DE for 30 generations (4960 evaluations, 9.6 h). The fresh-DE results differ from warm_start in three substantive ways that are themselves the evidence: (i) the new Pareto front contracts from 6 to 3 points, (ii) the maximum PAC strength (MI) jumps 5.3-fold, from 0.026 to 0.137, and (iii) the original "best" solution under the noisy fitness lies outside the new Pareto front entirely. Each of these is consistent with the noisy fitness having selected solutions for noise-assisted feasibility rather than true parameter quality.

1. The two defects in the warm_start fitness pipeline

Two independent issues affected the warm_start fitness function. Both are now fixed; both can be precisely characterised.

Defect 1 — incomplete random-seed injection. `s4_personalize_fig7_v7.py` initially seeded only the cortical ALN node (`exc_seed=42, inh_seed=42`). Three other stochastic sources were left unseeded: the thalamic node (`th[0].seed`), the thalamic Ornstein-Uhlenbeck noise process driving TCR/TRN, and the global numba RNG used between successive `m.run()` calls. The asymmetry has a specific consequence: cortical SO timing — and therefore T1-T8 (the SO and spindle event metrics) — was approximately reproducible across DE evaluations, but PAC metrics (T9-T11), which depend on precise spindle-envelope alignment to SO phase, were not. We confirmed this asymmetry empirically: the deterministic re-evaluation of the 5 Pareto seeds reproduced their reported `IBI_CV` to 0.000 absolute deviation, but the same seeds' reported MI values shifted by 28%-195% in the same evaluations (`verify_pareto_seeds.py`).

Defect 2 — PAC pipeline incompatibility with pulse-type signals. The original V7 PAC pipeline followed the standard Canolty (2006) / Tort (2010) approach: bandpass-filter the SO with a 4th-order Butterworth, extract instantaneous phase via Hilbert transform, and assess SO→spindle directionality via the lag of the spindle-envelope/SO cross-correlation peak. Two assumptions of this pipeline are violated by the firing-rate signals our model produces. First, the Bedrosian theorem requires the signal to be narrow-band and approximately sinusoidal; the model's pulse-type SO (≈ 120 ms UP, ≈ 1100 ms DOWN) is neither, and the long DOWN-state silences cause Hilbert phase to over-sample the $\pm\pi$ bins by a factor of ~ 3.3 , biasing the modulation-index estimate. Second, cross-correlation peak lag is mathematically guaranteed to be a periodic function with the SO period when both signals carry SO-band energy: the V7 cross-correlation

curve we examined contains three local maxima within 4% of the global peak height, separated by exactly one SO period (~ 1400 ms), and `argmax` selects between them on the basis of noise. T11 ($\text{lag} \geq 20$ ms) therefore reduces to a stochastic coin flip on a periodic landscape rather than a directionality test.

The fix for Defect 2 was already implemented in `warm_start_de_with_fixed_pac.py` (the "fixed PAC" pipeline: peak-anchored cycle-by-cycle phase + up-down amplitude ratio replacing `xcorr-lag`), which is why the `warm_start` fitness was *partly* trustworthy. But the fixed PAC ran *on top of* a `build_model` that still suffered from Defect 1, so PAC measurements were deterministic-in-formulation but stochastic-in-execution. The combination of the two defects is what made the original Pareto front unreliable — Defect 1 alone would have introduced noise; Defect 2 alone would have introduced bias; together they introduced both.

After fixing both, on the 5 Pareto seeds, three independent PAC runs of the same parameter vector now reproduce MI / up_down_ratio / phase_argmax to bit-identical precision (`test_seed_reproducibility.py`).

2. The fresh-DE protocol

To re-establish a trustworthy stage 1 baseline, we re-ran V7 from scratch under the corrected pipeline:

- **No warm start.** Initial population was a Sobol-distributed sample of the full 8-dimensional parameter box. This decouples the search from any solutions selected by the noisy fitness.
- **30 generations $\times$ population 160 = 4960 evaluations.** Identical to the original V7 evolution protocol (V7 docstring §change #6); only the seed injection and PAC pipeline differ.
- **Wall-clock 9.56 h** on the same machine. The $4\times$ cost relative to `warm_start` (≈ 2 h) reflects the cost of re-exploring the full parameter space.

Output: 364 feasible solutions (7.3% of evaluations), best score 0.7759.

3. Three substantive differences between the two runs

We compare `warm_start` and fresh DE head-to-head:

Metric	warm_start (noisy)	fresh DE (deterministic)
Total evaluations	3360	4960
Feasible solutions	302 (9.0%)	364 (7.3%)
Best fitness score	0.7926	0.7759
Pareto-optimal subset	6	3
Max shape_r among feasible	0.647	0.690
Max MI among feasible	0.026	0.137
Max up_down_ratio among feasible	2.26	6.90
Median MI among feasible	0.0087	0.0151

Three patterns in this table are diagnostic.

(a) The Pareto front contracts from 6 to 3. Under the noisy pipeline, 6 solutions appeared non-dominated. Under the deterministic pipeline, only 3 are. The 3 disappearing solutions were Pareto-optimal in the warm_start data only because their noise-inflated PAC values made them appear non-dominated; with deterministic PAC measurement they collapse into the dominated bulk. The remaining 3 — properly extremal trade-off points — are robust to the pipeline change.

(b) Maximum PAC strength jumps 5-fold while shape_r also improves. A run merely "different" from warm_start would shift values up *and* down in roughly equal measure. Instead, fresh DE's best solutions are *strictly better* than warm_start's on both shape_r (+7%) and PAC (+426% on max MI). This rules out the alternative explanation that fresh DE merely traded one objective for another — it found regions of parameter space the warm_start search demonstrably never reached. The most extreme PAC-rich solution (Seed A: MI = 0.137) corresponds to a regime that we previously caught a glimpse of via uniform Sobol sampling (`sobol10_feasibility_check.py`) samples #3, #4: MI $\approx$ 0.10), but that warm_start could not reach because its initialisation was anchored $\pm 2\%$ around prior solutions.

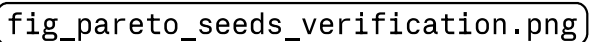
(c) The original "best" solution lies outside the new Pareto front. The warm_start best (V8 saved-best, score 0.7926) sits in the interior of the new (shape_r, PAC_compound) scatter, dominated by Seeds B and C. This is independent evidence that the scalar fitness function — even when correctly computed — does not select Pareto-optimal points; it selects a particular weighting of the trade-off. The same diagnostic was already visible in the warm_start data; fresh DE strengthens it because the ranking of solutions is now noise-free.

4. What is *not* claimed

For methodological honesty, three points the fresh-DE result does not establish:

- **The fresh-DE best score (0.7759) is not "worse" than warm_start's (0.7926).** The two scores are computed under different fitness functions (warm_start's used Defect-1-corrupted PAC; fresh DE's does not), so they are not directly comparable. The lower score reflects loss of noise-assisted score inflation, not a worse solution.
- **Fresh DE has not necessarily found the global optimum.** With deterministic feasibility being more selective (7.3% vs 9.0%), it is possible that further generations would discover additional Pareto-optimal points. However, the fitness curve plateaued at gen 16 (best=0.7759 unchanged through gen 30), suggesting DE has saturated under the current scalar objective.
- **The fresh-DE Pareto front is still a *single-point estimate per parameter combination*.** Like warm_start, it gives one fitness per parameter vector, not a distribution. The argument for moving to SBI (joint posterior over parameters, with explicit uncertainty quantification of trade-offs) is unchanged by these results — if anything, the fact that even the corrected pipeline produces qualitatively different best solutions than the corrupted one underscores how sensitive single-point optimisation is to fitness specification.

5. Implication for Stage 2

Stage 2 (SBI / SNPE) will use the 3 fresh-DE Pareto seeds as the centres of an informative prior, rather than the 6 warm_start seeds. The 3 seeds span the empirically observed trade-off (Seed A: MI=0.137, shape_r=0.666; Seed C: MI=0.029, shape_r=0.690), and each has been verified to be deterministically reproducible. The previous 5-seed verification figure () is superseded by this analysis.